by_normal_form_q11 over GF(11)

June 30, 2026

The equation

The equation of the quartic curve is :

$$X_0^4 + 4X_1^4 + 4X_2^4 + 5X_0^3X_1 + 4X_0^3X_2 + 4X_0X_1^3 + 4X_1^3X_2 + 3X_0X_2^3 \\ +7X_1X_2^3 + 4X_0^2X_2^2 + 4X_1^2X_2^2 + 10X_0^2X_1X_2 + X_0X_1^2X_2 + X_0X_1X_2^2$$

$$(1, 4, 4, 5, 4, 4, 4, 3, 7, 0, 4, 4, 10, 1, 1)$$

```
X0^4 + 4*X1^4 + 4*X2^4 + 5*X0^3*X1 + 4*X0^3*X2 + 4*X0*X1^3 + 4*X1^3*X2 + 3*X0*X2^3 + 7*X1*X2
```

General information

Number of bitangents	28
Number of points	0
Fullness	is full
Number of Kovalevski points	21
Bitangent line type (a_0, a_1, a_2)	$(28, 0, 0)$
Number of singular points	0

All points on the curve

The quartic has 0 points:
The points on the quartic curve are:

The points by rank are: ( )
The Kovalevski points are:
$0: P_{125} = (3, 10, 1) = b_2 \cap b_5 \cap c_{23} \cap c_{35}$
$1: P_{61} = (5, 4, 1) = a_2 \cap a_6 \cap c_{24} \cap c_{46}$
$2: P_{122} = (0, 10, 1) = a_2 \cap a_3 \cap b_2 \cap b_3$
$3: P_{120} = (9, 9, 1) = c_{24} \cap c_{26} \cap c_{34} \cap c_{36}$

$4:\ P_{28} = (5,1,1) = c_{12} \cap c_{13} \cap c_{25} \cap c_{35}$
$5:\ P_{109} = (9,8,1) = a_2 \cap b_6 \cap c_{15} \cap c_{34}$
$6:\ P_{26} = (3,1,1) = a_1 \cap a_6 \cap c_{12} \cap c_{26}$
$7:\ P_{103} = (3,8,1) = c_{14} \cap c_{15} \cap c_{46} \cap c_{56}$
$8:\ P_{51} = (6,3,1) = c_{16} \cap c_{23} \cap c_{45} \cap d$
$9:\ P_{101} = (1,8,1) = b_1 \cap b_3 \cap c_{15} \cap c_{35}$
$10:\ P_{11} = (8,1,0) = a_4 \cap a_6 \cap c_{14} \cap c_{16}$
$11:\ P_{94} = (5,7,1) = b_1 \cap b_5 \cap c_{16} \cap c_{56}$
$12:\ P_{21} = (8,0,1) = b_4 \cap b_6 \cap c_{45} \cap c_{56}$
$13:\ P_{10} = (7,1,0) = b_1 \cap b_6 \cap c_{13} \cap c_{36}$
$14:\ P_{78} = (0,6,1) = a_3 \cap a_4 \cap c_{23} \cap c_{24}$
$15:\ P_{35} = (1,2,1) = a_4 \cap b_5 \cap c_{13} \cap c_{26}$
$16:\ P_{8} = (5,1,0) = a_1 \cap b_3 \cap c_{25} \cap c_{46}$
$17:\ P_{68} = (1,5,1) = a_1 \cap a_5 \cap c_{14} \cap c_{45}$
$18:\ P_{128} = (6,10,1) = a_5 \cap b_2 \cap c_{25} \cap d$
$19:\ P_{31} = (8,1,1) = a_5 \cap b_4 \cap c_{12} \cap c_{36}$
$20:\ P_{1} = (0,1,0) = a_3 \cap b_4 \cap c_{34} \cap d$
The incidence structure induced by the Kovalevski points is:
Flags=(6, 16, 17, 22, 23, 26, 44, 56, 62, 73, 77, 78, 101, 102, 103, 106, 111, 115, 135, 137, 139, 147, 149, 165, 170, 177, 184, 201, 208, 209, 210, 221, 225, 236, 243, 244, 256, 258, 271, 277, 286, 288, 301, 304, 311, 320, 322, 324, 344, 346, 347, 357, 365, 371, 379, 381, 392, 403, 415, 417, 423, 426, 435, 444, 446, 461, 462, 466, 471, 486, 496, 502, 512, 516, 521, 526, 532, 541, 553, 557, 558, 575, 585, 587)
nb_rows=28
nb_cols=21
nb_flags=84
Incma=
(0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0)
(0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
(0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0)
(0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0)
(1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0)
(0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 1)
(1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0)
(0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0)
(0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0)
(1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)

(0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0)
(0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)
(0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1)
(1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0)
(0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1)
The Kovalevski points sorted by rank are: (1, 8, 10, 11, 21, 26, 28, 31, 35, 51, 61, 68, 78, 94, 101, 103, 109, 120, 122, 125, 128)
The points off the curve are:

0 : $P_0 = (1,0,0)$
1 : $P_1 = (0,1,0)$
2 : $P_2 = (0,0,1)$
3 : $P_3 = (1,1,1)$
4 : $P_4 = (1,1,0)$
5 : $P_5 = (2,1,0)$
6 : $P_6 = (3,1,0)$
7 : $P_7 = (4,1,0)$
8 : $P_8 = (5,1,0)$
9 : $P_9 = (6,1,0)$
10 : $P_{10} = (7,1,0)$
11 : $P_{11} = (8,1,0)$
12 : $P_{12} = (9,1,0)$
13 : $P_{13} = (10,1,0)$
14 : $P_{14} = (1,0,1)$
15 : $P_{15} = (2,0,1)$
16 : $P_{16} = (3,0,1)$
17 : $P_{17} = (4,0,1)$
18 : $P_{18} = (5,0,1)$
19 : $P_{19} = (6,0,1)$
20 : $P_{20} = (7,0,1)$
21 : $P_{21} = (8,0,1)$
22 : $P_{22} = (9,0,1)$
23 : $P_{23} = (10,0,1)$
24 : $P_{24} = (0,1,1)$
25 : $P_{25} = (2,1,1)$
26 : $P_{26} = (3,1,1)$
27 : $P_{27} = (4,1,1)$
28 : $P_{28} = (5,1,1)$
29 : $P_{29} = (6,1,1)$
30 : $P_{30} = (7,1,1)$
31 : $P_{31} = (8,1,1)$
32 : $P_{32} = (9,1,1)$
33 : $P_{33} = (10,1,1)$
34 : $P_{34} = (0,2,1)$
35 : $P_{35} = (1,2,1)$
36 : $P_{36} = (2,2,1)$
37 : $P_{37} = (3,2,1)$
38 : $P_{38} = (4,2,1)$
39 : $P_{39} = (5,2,1)$
40 : $P_{40} = (6,2,1)$
41 : $P_{41} = (7,2,1)$
42 : $P_{42} = (8,2,1)$
43 : $P_{43} = (9,2,1)$
44 : $P_{44} = (10,2,1)$
45 : $P_{45} = (0,3,1)$
46 : $P_{46} = (1,3,1)$
47 : $P_{47} = (2,3,1)$
48 : $P_{48} = (3,3,1)$
49 : $P_{49} = (4,3,1)$
50 : $P_{50} = (5,3,1)$
51 : $P_{51} = (6,3,1)$
52 : $P_{52} = (7,3,1)$
53 : $P_{53} = (8,3,1)$
54 : $P_{54} = (9,3,1)$
55 : $P_{55} = (10,3,1)$
56 : $P_{56} = (0,4,1)$
57 : $P_{57} = (1,4,1)$
58 : $P_{58} = (2,4,1)$
59 : $P_{59} = (3,4,1)$
60 : $P_{60} = (4,4,1)$
61 : $P_{61} = (5,4,1)$
62 : $P_{62} = (6,4,1)$
63 : $P_{63} = (7,4,1)$
64 : $P_{64} = (8,4,1)$
65 : $P_{65} = (9,4,1)$
66 : $P_{66} = (10,4,1)$
67 : $P_{67} = (0,5,1)$
68 : $P_{68} = (1,5,1)$
69 : $P_{69} = (2,5,1)$
70 : $P_{70} = (3,5,1)$
71 : $P_{71} = (4,5,1)$
72 : $P_{72} = (5,5,1)$
73 : $P_{73} = (6,5,1)$
74 : $P_{74} = (7,5,1)$
75 : $P_{75} = (8,5,1)$
76 : $P_{76} = (9,5,1)$
77 : $P_{77} = (10,5,1)$
78 : $P_{78} = (0,6,1)$
79 : $P_{79} = (1,6,1)$
80 : $P_{80} = (2,6,1)$
81 : $P_{81} = (3,6,1)$
82 : $P_{82} = (4,6,1)$
83 : $P_{83} = (5,6,1)$
84 : $P_{84} = (6,6,1)$
85 : $P_{85} = (7,6,1)$
86 : $P_{86} = (8,6,1)$
87 : $P_{87} = (9,6,1)$
88 : $P_{88} = (10,6,1)$
89 : $P_{89} = (0,7,1)$
90 : $P_{90} = (1,7,1)$
91 : $P_{91} = (2,7,1)$
92 : $P_{92} = (3,7,1)$

$93: P_{93} = (4,7,1)$	$107: P_{107} = (7,8,1)$	$121: P_{121} = (10,9,1)$
$94: P_{94} = (5,7,1)$	$108: P_{108} = (8,8,1)$	$122: P_{122} = (0,10,1)$
$95: P_{95} = (6,7,1)$	$109: P_{109} = (9,8,1)$	$123: P_{123} = (1,10,1)$
$96: P_{96} = (7,7,1)$	$110: P_{110} = (10,8,1)$	$124: P_{124} = (2,10,1)$
$97: P_{97} = (8,7,1)$	$111: P_{111} = (0,9,1)$	$125: P_{125} = (3,10,1)$
$98: P_{98} = (9,7,1)$	$112: P_{112} = (1,9,1)$	$126: P_{126} = (4,10,1)$
$99: P_{99} = (10,7,1)$	$113: P_{113} = (2,9,1)$	$127: P_{127} = (5,10,1)$
$100: P_{100} = (0,8,1)$	$114: P_{114} = (3,9,1)$	$128: P_{128} = (6,10,1)$
$101: P_{101} = (1,8,1)$	$115: P_{115} = (4,9,1)$	$129: P_{129} = (7,10,1)$
$102: P_{102} = (2,8,1)$	$116: P_{116} = (5,9,1)$	$130: P_{130} = (8,10,1)$
$103: P_{103} = (3,8,1)$	$117: P_{117} = (6,9,1)$	$131: P_{131} = (9,10,1)$
$104: P_{104} = (4,8,1)$	$118: P_{118} = (7,9,1)$	$132: P_{132} = (10,10,1)$
$105: P_{105} = (5,8,1)$	$119: P_{119} = (8,9,1)$	
$106: P_{106} = (6,8,1)$	$120: P_{120} = (9,9,1)$	

(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132)

Duals of Bitangents

dual of bitangents:

$0: P_{51} = (6,3,1)$
$1: P_{33} = (10,1,1)$
$2: P_{0} = (1,0,0)$
$3: P_{114} = (3,9,1)$
$4: P_{103} = (3,8,1)$
$5: P_{76} = (9,5,1)$
$6: P_{6} = (3,1,0)$
$7: P_{24} = (0,1,1)$
$8: P_{25} = (2,1,1)$
$9: P_{17} = (4,0,1)$
$10: P_{87} = (9,6,1)$
$11: P_{71} = (4,5,1)$
$12: P_{122} = (0,10,1)$
$13: P_{40} = (6,2,1)$
$14: P_{80} = (2,6,1)$
$15: P_{56} = (0,4,1)$
$16: P_{42} = (8,2,1)$
$17: P_{121} = (10,9,1)$
$18: P_{119} = (8,9,1)$

$19 : P_5 = (2, 1, 0)$
$20 : P_{58} = (2, 4, 1)$
$21 : P_{19} = (6, 0, 1)$
$22 : P_{53} = (8, 3, 1)$
$23 : P_{99} = (10, 7, 1)$
$24 : P_{126} = (4, 10, 1)$
$25 : P_{92} = (3, 7, 1)$
$26 : P_{104} = (4, 8, 1)$
$27 : P_{22} = (9, 0, 1)$
Dimension of the vanishing ideal is 0

The lines and their points of contact are:
$a_1 = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 8 \end{bmatrix}_{68}$
$a_2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 10 \end{bmatrix}_{22}$
$a_3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{132}$
$a_4 = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 2 \end{bmatrix}_{98}$
$a_5 = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 3 \end{bmatrix}_{99}$
$a_6 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 6 \end{bmatrix}_{30}$
$b_1 = \begin{bmatrix} 1 & 8 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{107}$
$b_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 10 \end{bmatrix}_{10}$
$b_3 = \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 10 \end{bmatrix}_{118}$
$b_4 = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 0 \end{bmatrix}_{84}$
$b_5 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 5 \end{bmatrix}_{29}$
$b_6 = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 6 \end{bmatrix}_{90}$
$c_{12} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}_{1}$
$c_{13} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 9 \end{bmatrix}_{69}$
$c_{14} = \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 5 \end{bmatrix}_{113}$

$$c_{15} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 7 \end{bmatrix}_{7}$$
$$c_{16} = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 9 \end{bmatrix}_{45}$$
$$c_{23} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}_{14}$$
$$c_{24} = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix}_{38}$$
$$c_{25} = \begin{bmatrix} 1 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{119}$$
$$c_{26} = \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 7 \end{bmatrix}_{115}$$
$$c_{34} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 0 \end{bmatrix}_{60}$$
$$c_{35} = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 8 \end{bmatrix}_{44}$$
$$c_{36} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 4 \end{bmatrix}_{16}$$
$$c_{45} = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 1 \end{bmatrix}_{85}$$
$$c_{46} = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 4 \end{bmatrix}_{100}$$
$$c_{56} = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 3 \end{bmatrix}_{87}$$
$$d = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \end{bmatrix}_{24}$$

Rank of lines: (68, 22, 132, 98, 99, 30, 107, 10, 118, 84, 29, 90, 1, 69, 113, 7, 45, 14, 38, 119, 115, 60, 44, 16, 85, 100, 87, 24)

Line type: 0^{28}

28	0	$1, 2, 3, 4, 5, 6, 7, 8, 9, 10,$ $11, 12, 13, 14, 15, 16, 17, 18, 19, 20,$ $21, 22, 23, 24, 25, 26, 27, 0$

point types:

point types for points off the curve: 4^{21}, 3^{84}, 0^{28}

21	4	$125, 61, 122, 120, 28, 109, 26, 103, 51, 101, 11, 94, 21, 10, 78, 35, 8, 68, 128, 31, 1$
84	3	$30, 14, 29, 116, 55, 27, 6, 121, 123, 59, 49, 47, 2, 119, 113, 22, 91, 45, 95, 97, 100, 24, 12, 52, 53, 60, 111, 56, 44, 117, 43, 4, 126, 42, 84, 20, 40, 80, 19, 62, 18, 37, 130, 73, 17, 131, 75, 76, 77, 38, 129, 70, 69, 82, 34, 33, 85, 86, 87, 88, 89, 67, 0, 92, 65, 32, 64, 96, 63, 98, 127, 3, 132, 102, 25, 104, 105, 106, 107, 108, 54, 110, 124, 112$
28	0	$58, 46, 15, 115, 74, 99, 83, 71, 114, 118, 16, 36, 90, 50, 48, 41, 79, 72, 23, 5, 57, 13, 7, 66, 93, 9, 39, 81$

Lines on points off the curve:
Off point 0 = $P_0 = (1, 0, 0)$ lies on 3 bisecants : { 7, 12, 15 }
Off point 1 = $P_1 = (0, 1, 0)$ lies on 4 bisecants : { 2, 9, 21, 27 }
Off point 2 = $P_2 = (0, 0, 1)$ lies on 3 bisecants : { 2, 6, 19 }
Off point 3 = $P_3 = (1, 1, 1)$ lies on 3 bisecants : { 12, 16, 25 }
Off point 4 = $P_4 = (1, 1, 0)$ lies on 3 bisecants : { 1, 4, 22 }
Off point 5 = $P_5 = (2, 1, 0)$ lies on 0 bisecants : { }
Off point 6 = $P_6 = (3, 1, 0)$ lies on 3 bisecants : { 10, 18, 24 }
Off point 7 = $P_7 = (4, 1, 0)$ lies on 0 bisecants : { }
Off point 8 = $P_8 = (5, 1, 0)$ lies on 4 bisecants : { 0, 8, 19, 25 }
Off point 9 = $P_9 = (6, 1, 0)$ lies on 0 bisecants : { }
Off point 10 = $P_{10} = (7, 1, 0)$ lies on 4 bisecants : { 6, 11, 13, 23 }
Off point 11 = $P_{11} = (8, 1, 0)$ lies on 4 bisecants : { 3, 5, 14, 16 }
Off point 12 = $P_{12} = (9, 1, 0)$ lies on 3 bisecants : { 17, 20, 26 }
Off point 13 = $P_{13} = (10, 1, 0)$ lies on 0 bisecants : { }
Off point 14 = $P_{14} = (1, 0, 1)$ lies on 3 bisecants : { 1, 17, 23 }
Off point 15 = $P_{15} = (2, 0, 1)$ lies on 0 bisecants : { }
Off point 16 = $P_{16} = (3, 0, 1)$ lies on 0 bisecants : { }
Off point 17 = $P_{17} = (4, 0, 1)$ lies on 3 bisecants : { 16, 18, 22 }
Off point 18 = $P_{18} = (5, 0, 1)$ lies on 3 bisecants : { 8, 14, 20 }
Off point 19 = $P_{19} = (6, 0, 1)$ lies on 3 bisecants : { 5, 10, 27 }
Off point 20 = $P_{20} = (7, 0, 1)$ lies on 3 bisecants : { 3, 4, 25 }
Off point 21 = $P_{21} = (8, 0, 1)$ lies on 4 bisecants : { 9, 11, 24, 26 }
Off point 22 = $P_{22} = (9, 0, 1)$ lies on 3 bisecants : { 0, 13, 21 }
Off point 23 = $P_{23} = (10, 0, 1)$ lies on 0 bisecants : { }
Off point 24 = $P_{24} = (0, 1, 1)$ lies on 3 bisecants : { 2, 12, 24 }
Off point 25 = $P_{25} = (2, 1, 1)$ lies on 3 bisecants : { 1, 12, 14 }
Off point 26 = $P_{26} = (3, 1, 1)$ lies on 4 bisecants : { 0, 5, 12, 20 }

Off point 27 $= P_{27} = (4,1,1)$ lies on 3 bisecants : { 3, 11, 12 }
Off point 28 $= P_{28} = (5,1,1)$ lies on 4 bisecants : { 12, 13, 19, 22 }
Off point 29 $= P_{29} = (6,1,1)$ lies on 3 bisecants : { 12, 26, 27 }
Off point 30 $= P_{30} = (7,1,1)$ lies on 3 bisecants : { 6, 12, 18 }
Off point 31 $= P_{31} = (8,1,1)$ lies on 4 bisecants : { 4, 9, 12, 23 }
Off point 32 $= P_{32} = (9,1,1)$ lies on 3 bisecants : { 10, 12, 21 }
Off point 33 $= P_{33} = (10,1,1)$ lies on 3 bisecants : { 8, 12, 17 }
Off point 34 $= P_{34} = (0,2,1)$ lies on 3 bisecants : { 2, 5, 11 }
Off point 35 $= P_{35} = (1,2,1)$ lies on 4 bisecants : { 3, 10, 13, 20 }
Off point 36 $= P_{36} = (2,2,1)$ lies on 0 bisecants : { }
Off point 37 $= P_{37} = (3,2,1)$ lies on 3 bisecants : { 1, 6, 24 }
Off point 38 $= P_{38} = (4,2,1)$ lies on 3 bisecants : { 8, 23, 26 }
Off point 39 $= P_{39} = (5,2,1)$ lies on 0 bisecants : { }
Off point 40 $= P_{40} = (6,2,1)$ lies on 3 bisecants : { 22, 25, 27 }
Off point 41 $= P_{41} = (7,2,1)$ lies on 0 bisecants : { }
Off point 42 $= P_{42} = (8,2,1)$ lies on 3 bisecants : { 0, 9, 17 }
Off point 43 $= P_{43} = (9,2,1)$ lies on 3 bisecants : { 4, 16, 21 }
Off point 44 $= P_{44} = (10,2,1)$ lies on 3 bisecants : { 14, 18, 19 }
Off point 45 $= P_{45} = (0,3,1)$ lies on 3 bisecants : { 2, 23, 25 }
Off point 46 $= P_{46} = (1,3,1)$ lies on 0 bisecants : { }
Off point 47 $= P_{47} = (2,3,1)$ lies on 3 bisecants : { 0, 18, 26 }
Off point 48 $= P_{48} = (3,3,1)$ lies on 0 bisecants : { }
Off point 49 $= P_{49} = (4,3,1)$ lies on 3 bisecants : { 1, 10, 19 }
Off point 50 $= P_{50} = (5,3,1)$ lies on 0 bisecants : { }
Off point 51 $= P_{51} = (6,3,1)$ lies on 4 bisecants : { 16, 17, 24, 27 }
Off point 52 $= P_{52} = (7,3,1)$ lies on 3 bisecants : { 11, 14, 22 }
Off point 53 $= P_{53} = (8,3,1)$ lies on 3 bisecants : { 5, 9, 13 }
Off point 54 $= P_{54} = (9,3,1)$ lies on 3 bisecants : { 3, 8, 21 }
Off point 55 $= P_{55} = (10,3,1)$ lies on 3 bisecants : { 4, 6, 20 }
Off point 56 $= P_{56} = (0,4,1)$ lies on 3 bisecants : { 2, 4, 26 }
Off point 57 $= P_{57} = (1,4,1)$ lies on 0 bisecants : { }
Off point 58 $= P_{58} = (2,4,1)$ lies on 0 bisecants : { }
Off point 59 $= P_{59} = (3,4,1)$ lies on 3 bisecants : { 8, 11, 16 }
Off point 60 $= P_{60} = (4,4,1)$ lies on 3 bisecants : { 13, 14, 17 }
Off point 61 $= P_{61} = (5,4,1)$ lies on 4 bisecants : { 1, 5, 18, 25 }
Off point 62 $= P_{62} = (6,4,1)$ lies on 3 bisecants : { 3, 6, 27 }
Off point 63 $= P_{63} = (7,4,1)$ lies on 3 bisecants : { 0, 10, 23 }
Off point 64 $= P_{64} = (8,4,1)$ lies on 3 bisecants : { 9, 20, 22 }
Off point 65 $= P_{65} = (9,4,1)$ lies on 3 bisecants : { 19, 21, 24 }
Off point 66 $= P_{66} = (10,4,1)$ lies on 0 bisecants : { }
Off point 67 $= P_{67} = (0,5,1)$ lies on 3 bisecants : { 2, 13, 16 }
Off point 68 $= P_{68} = (1,5,1)$ lies on 4 bisecants : { 0, 4, 14, 24 }
Off point 69 $= P_{69} = (2,5,1)$ lies on 3 bisecants : { 5, 6, 17 }
Off point 70 $= P_{70} = (3,5,1)$ lies on 3 bisecants : { 3, 19, 23 }
Off point 71 $= P_{71} = (4,5,1)$ lies on 0 bisecants : { }
Off point 72 $= P_{72} = (5,5,1)$ lies on 0 bisecants : { }

Off point 73 = $P_{73} = (6, 5, 1)$ lies on 3 bisecants : { 1, 20, 27 }
Off point 74 = $P_{74} = (7, 5, 1)$ lies on 0 bisecants : { }
Off point 75 = $P_{75} = (8, 5, 1)$ lies on 3 bisecants : { 8, 9, 18 }
Off point 76 = $P_{76} = (9, 5, 1)$ lies on 3 bisecants : { 21, 22, 26 }
Off point 77 = $P_{77} = (10, 5, 1)$ lies on 3 bisecants : { 10, 11, 25 }
Off point 78 = $P_{78} = (0, 6, 1)$ lies on 4 bisecants : { 2, 3, 17, 18 }
Off point 79 = $P_{79} = (1, 6, 1)$ lies on 0 bisecants : { }
Off point 80 = $P_{80} = (2, 6, 1)$ lies on 3 bisecants : { 4, 8, 10 }
Off point 81 = $P_{81} = (3, 6, 1)$ lies on 0 bisecants : { }
Off point 82 = $P_{82} = (4, 6, 1)$ lies on 3 bisecants : { 20, 24, 25 }
Off point 83 = $P_{83} = (5, 6, 1)$ lies on 0 bisecants : { }
Off point 84 = $P_{84} = (6, 6, 1)$ lies on 3 bisecants : { 0, 11, 27 }
Off point 85 = $P_{85} = (7, 6, 1)$ lies on 3 bisecants : { 1, 13, 26 }
Off point 86 = $P_{86} = (8, 6, 1)$ lies on 3 bisecants : { 9, 16, 19 }
Off point 87 = $P_{87} = (9, 6, 1)$ lies on 3 bisecants : { 6, 14, 21 }
Off point 88 = $P_{88} = (10, 6, 1)$ lies on 3 bisecants : { 5, 22, 23 }
Off point 89 = $P_{89} = (0, 7, 1)$ lies on 3 bisecants : { 0, 2, 22 }
Off point 90 = $P_{90} = (1, 7, 1)$ lies on 0 bisecants : { }
Off point 91 = $P_{91} = (2, 7, 1)$ lies on 3 bisecants : { 11, 19, 20 }
Off point 92 = $P_{92} = (3, 7, 1)$ lies on 3 bisecants : { 4, 13, 18 }
Off point 93 = $P_{93} = (4, 7, 1)$ lies on 0 bisecants : { }
Off point 94 = $P_{94} = (5, 7, 1)$ lies on 4 bisecants : { 6, 10, 16, 26 }
Off point 95 = $P_{95} = (6, 7, 1)$ lies on 3 bisecants : { 14, 23, 27 }
Off point 96 = $P_{96} = (7, 7, 1)$ lies on 3 bisecants : { 5, 8, 24 }
Off point 97 = $P_{97} = (8, 7, 1)$ lies on 3 bisecants : { 1, 3, 9 }
Off point 98 = $P_{98} = (9, 7, 1)$ lies on 3 bisecants : { 17, 21, 25 }
Off point 99 = $P_{99} = (10, 7, 1)$ lies on 0 bisecants : { }
Off point 100 = $P_{100} = (0, 8, 1)$ lies on 3 bisecants : { 2, 15, 20 }
Off point 101 = $P_{101} = (1, 8, 1)$ lies on 4 bisecants : { 6, 8, 15, 22 }
Off point 102 = $P_{102} = (2, 8, 1)$ lies on 3 bisecants : { 15, 16, 23 }
Off point 103 = $P_{103} = (3, 8, 1)$ lies on 4 bisecants : { 14, 15, 25, 26 }
Off point 104 = $P_{104} = (4, 8, 1)$ lies on 3 bisecants : { 4, 5, 15 }
Off point 105 = $P_{105} = (5, 8, 1)$ lies on 3 bisecants : { 0, 3, 15 }
Off point 106 = $P_{106} = (6, 8, 1)$ lies on 3 bisecants : { 15, 18, 27 }
Off point 107 = $P_{107} = (7, 8, 1)$ lies on 3 bisecants : { 15, 17, 19 }
Off point 108 = $P_{108} = (8, 8, 1)$ lies on 3 bisecants : { 9, 10, 15 }
Off point 109 = $P_{109} = (9, 8, 1)$ lies on 4 bisecants : { 1, 11, 15, 21 }
Off point 110 = $P_{110} = (10, 8, 1)$ lies on 3 bisecants : { 13, 15, 24 }
Off point 111 = $P_{111} = (0, 9, 1)$ lies on 3 bisecants : { 2, 10, 14 }
Off point 112 = $P_{112} = (1, 9, 1)$ lies on 3 bisecants : { 5, 19, 26 }
Off point 113 = $P_{113} = (2, 9, 1)$ lies on 3 bisecants : { 3, 22, 24 }
Off point 114 = $P_{114} = (3, 9, 1)$ lies on 0 bisecants : { }
Off point 115 = $P_{115} = (4, 9, 1)$ lies on 0 bisecants : { }
Off point 116 = $P_{116} = (5, 9, 1)$ lies on 3 bisecants : { 4, 11, 17 }
Off point 117 = $P_{117} = (6, 9, 1)$ lies on 3 bisecants : { 8, 13, 27 }
Off point 118 = $P_{118} = (7, 9, 1)$ lies on 0 bisecants : { }

Off point 119 $= P_{119} = (8,9,1)$ lies on 3 bisecants : { 6, 9, 25 }
Off point 120 $= P_{120} = (9,9,1)$ lies on 4 bisecants : { 18, 20, 21, 23 }
Off point 121 $= P_{121} = (10,9,1)$ lies on 3 bisecants : { 0, 1, 16 }
Off point 122 $= P_{122} = (0,10,1)$ lies on 4 bisecants : { 1, 2, 7, 8 }
Off point 123 $= P_{123} = (1,10,1)$ lies on 3 bisecants : { 7, 11, 18 }
Off point 124 $= P_{124} = (2,10,1)$ lies on 3 bisecants : { 7, 13, 25 }
Off point 125 $= P_{125} = (3,10,1)$ lies on 4 bisecants : { 7, 10, 17, 22 }
Off point 126 $= P_{126} = (4,10,1)$ lies on 3 bisecants : { 0, 6, 7 }
Off point 127 $= P_{127} = (5,10,1)$ lies on 3 bisecants : { 7, 23, 24 }
Off point 128 $= P_{128} = (6,10,1)$ lies on 4 bisecants : { 4, 7, 19, 27 }
Off point 129 $= P_{129} = (7,10,1)$ lies on 3 bisecants : { 7, 16, 20 }
Off point 130 $= P_{130} = (8,10,1)$ lies on 3 bisecants : { 7, 9, 14 }
Off point 131 $= P_{131} = (9,10,1)$ lies on 3 bisecants : { 5, 7, 21 }
Off point 132 $= P_{132} = (10,10,1)$ lies on 3 bisecants : { 3, 7, 26 }

The gradient

The gradient of the quartic curve is :

$$4X_0^3 + 4X_1^3 + 3X_2^3 + 4X_0^2X_1 + X_0^2X_2 + X_1^2X_2 + 8X_0X_2^2 + X_1X_2^2 \\ +9X_0X_1X_2$$

$$(4,4,3,4,1,0,1,8,1,9)$$

$$5X_0^3 + 5X_1^3 + 7X_2^3 + 10X_0^2X_2 + X_0X_1^2 + X_1^2X_2 + X_0X_2^2 + 8X_1X_2^2 \\ +2X_0X_1X_2$$

$$(5,5,7,0,10,1,1,1,8,2)$$

$$4X_0^3 + 4X_1^3 + 5X_2^3 + 10X_0^2X_1 + 8X_0^2X_2 + X_0X_1^2 + 8X_1^2X_2 + 9X_0X_2^2 \\ +10X_1X_2^2 + 2X_0X_1X_2$$

$$(4,4,5,10,8,1,8,9,10,2)$$